

TAMING MULTI-AGENT COLLABORATION FOR 3D SITUATED REASONING VIA UNCERTAINTY ALIGNMENT

Yirong Qiang^{1,*} Xi Hong^{2,*} Zhewei Li³
Yuling Zheng¹ Yi Lu⁴ Yilun Chen^{5,†}

¹University of Electronic Science and Technology of China

²Harbin Institute of Technology

³Beihang University

⁴Huazhong University of Science and Technology

⁵Tsinghua University

ABSTRACT

Situated reasoning requires reconciling observations from an agent’s viewpoint with the geometry of an entire scene. Experts with complementary evidence can disagree, while their confidence scores may have incompatible scales. We introduce TAME3D, a framework that coordinates a manager, egocentric and global experts, and an evidence-checking critic. Typed geometric operations provide inspectable evidence, temperature scaling aligns expert scores, and scene-level conformal prediction governs answering, further acquisition, and abstention. An explicit residual category accounts for answers omitted from the candidate set. Under exchangeability and a frozen acquisition policy, we bound the probability of any incorrect automatically accepted answer in a new scene. Experiments on synthetic 3D data evaluate the uncertainty controller across five random seeds. Alignment improves forced accuracy by 2.00 percentage points. Adaptive inference achieves 97.39% selective accuracy with an answer rate of 83.60%, using 46.80% fewer additional acquisitions than a final fixed-round rule. Under observation shift, mapped-label coverage falls from 96.70% to 88.16%. These results characterize the tradeoff between acquisition cost, answer rate, and reliability, together with the limits of calibration under shift.

1 INTRODUCTION

Situated question answering links a language query to a particular position and orientation in a 3D environment. Answering “What is to my left?” requires identifying the reference frame, accounting for visibility, and grounding the requested relation in scene geometry. SQA3D makes this dependence explicit by pairing indoor scenes with situation descriptions and questions (Ma et al., 2022). Such tasks challenge foundation models to connect language with spatial evidence, particularly when a plausible answer rests on an incorrect pose or coordinate convention.

Existing approaches learn joint scene-language representations or introduce situation-aware encodings (Zhu et al., 2023; Man et al., 2024). Program-based systems instead expose intermediate operations that an external executor can check (He et al., 2024). Multi-agent VQA suggests a further division of labor between specialized visual and reasoning roles (Jiang et al., 2024). However, adding agents introduces a decision problem: two experts may disagree because one lacks visual evidence, because they use different coordinate conventions, or because both inherit the same detector error. A majority vote does not distinguish these cases, and self-reported confidence may reward assertive wording.

We propose TAME3D, an uncertainty-aware orchestration framework with four roles. A manager establishes the coordinate frame and acquisition plan. An egocentric expert reasons from local views, while a global expert queries scene geometry. A critic checks the evidence supporting their

*These authors contributed equally.

†Corresponding author.

candidate answers. A fixed scoring rule aligns and combines the checked outputs, and a conformal prediction set determines whether the evidence supports a single answer. Unresolved questions receive additional observations within a fixed budget, followed by abstention if ambiguity remains.

The framework separates two sources of uncertainty. Expert scores can have different scales, which we address with temperature scaling on development data. Candidate generation can also omit the correct answer, which we represent through an explicit residual category. A separate calibration split then determines whether to answer. This procedure uses labeled development and calibration scenes even when the underlying model weights remain frozen.

Our contributions are threefold: an evidence-based agent and tool interface for situated 3D reasoning; a candidate-aware conformal rule with a scene-level guarantee under bounded adaptive acquisition; and a controlled evaluation on synthetic 3D data that isolates score alignment, abstention, and observation shift. The experiments use geometric estimators with incomplete and noisy observations, allowing us to examine the uncertainty controller independently of learned perception and language generation.

2 RELATED WORK

Situated 3D reasoning. SQA3D evaluates reasoning conditioned on a situation in a reconstructed scene (Ma et al., 2022). 3D-VisTA pretrains a transformer for 3D vision-language alignment (Zhu et al., 2023), while SIG3D emphasizes situation awareness (Man et al., 2024). Think-Program-reCtify uses executable programs and feedback for situated reasoning (He et al., 2024). These works motivate explicit geometric operations; they do not by themselves establish a calibrated rule for choosing between conflicting expert answers.

Foundation-model collaboration. Multi-Agent VQA decomposes visual question answering among foundation-model agents (Jiang et al., 2024). Our decomposition assigns distinct evidence access to local and global roles and preserves object identifiers across their messages. We focus on deciding when the combined evidence warrants an answer and when another observation is needed. Because correlated experts can agree on an error, agreement alone is insufficient to establish correctness.

Uncertainty and selective prediction. Temperature scaling adjusts predictive scores using held-out labels (Guo et al., 2017). Split conformal prediction converts fixed nonconformity scores into prediction sets with marginal coverage under exchangeability (Angelopoulos & Bates, 2021). We apply these ideas to the fused output of an agent system. Our guarantee explicitly requires exchangeable scene blocks and a policy fixed before calibration. Set coverage also differs from the accuracy among answered questions; we report both quantities separately.

3 METHOD

3.1 TASK, EVIDENCE, AND COORDINATE FRAMES

Let $x = (\mathcal{S}, d, q)$ contain a scene, a situation description, and a question, with canonical answer $y \in \mathcal{Y}$. The scene interface exposes available RGB/depth views, object instances with stable identifiers, and metric geometry. A pose $T = (R, t)$, with $R \in \text{SO}(3)$ and $t \in \mathbb{R}^3$, maps local coordinates to world coordinates. Every spatial tool uses

$$v_{\text{ego}} = R^\top (v_{\text{world}} - t), \quad (1)$$

with $+x$ right, $+y$ forward, and $+z$ up. Dataset-specific axes are converted once at input. For example, an object is left of the agent if its ego-frame horizontal coordinate is negative beyond a fixed tolerance. The evidence record includes this frame, the pose source, object IDs, units, and visibility status. A relation inferred from an occluded object is marked as geometry-based evidence, not as a directly observed pixel fact.

We distinguish *oracle-pose* input, where the annotated location and orientation are provided, from *inferred-pose* input, where a frozen localizer estimates them from d and the scene. Results from

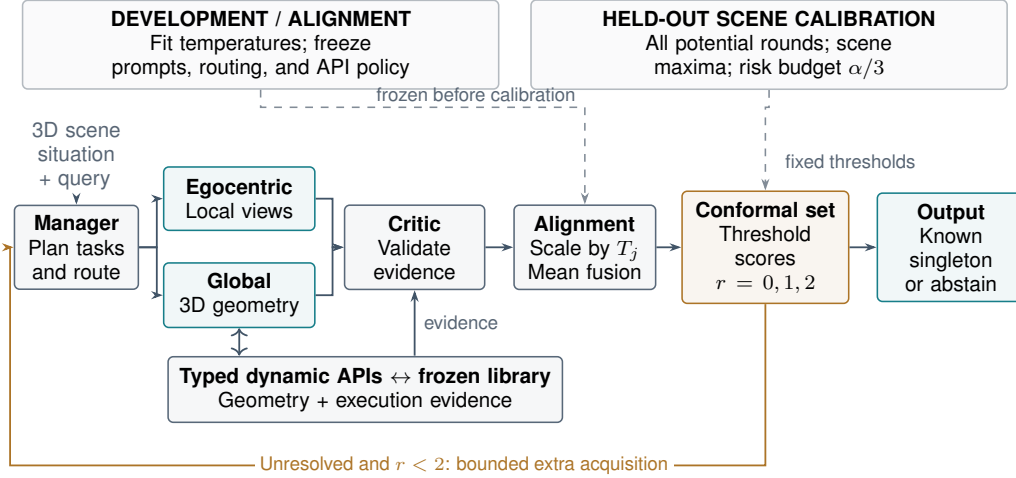


Figure 1: TAME3D architecture, illustrated for two additional acquisitions. Experts use complementary evidence with shared object identifiers and coordinate conventions. Temperatures are fitted on development scenes; thresholds are fitted on separate calibration scenes. The acquisition policy is frozen before calibration, and the thresholds determine only when to stop.

these regimes are not interchangeable. Pose uncertainty may be represented by a fixed number of hypotheses $\{T_h\}$ with nonnegative weights summing to one. Their scoring and aggregation rules are fixed on development data.

3.2 SPECIALIZED AGENTS AND A TYPED DYNAMIC API

The **manager** canonicalizes the question, records the pose regime, and follows an acquisition policy with a maximum of B additional rounds. The budget is fixed during development. At each potential round $r \in \{0, \dots, B\}$, the **egocentric expert** receives local views and visibility evidence; the **global expert** receives a scene graph and metric tools. Both score a shared candidate list and cite the evidence supporting their scores. The **critic** checks schema validity, frame consistency, object existence, and whether the cited operations support each relation. Its verdict enters a fixed score transformation before fusion. The critic cannot alter the candidates or replace the final answer after the conformal set is computed.

The dynamic API composes a typed vocabulary of detect, select, transform, visible, relate, count, and nearest. Each successful call returns a value and an evidence record. Type errors, unavailable geometry, and unresolved numerical boundaries produce explicit failure states. An empty detection is distinguished from verified absence, so a missed object cannot justify a count of zero. Programs are stored as normalized operation graphs with typed arguments and preconditions. Program templates can be reused across scenes; cached results require matching scene version, pose, view, arguments, and tool version.

The executor design restricts generated programs to this vocabulary, with bounded call count and execution time. Templates and repair rules are selected on development scenes, then frozen before calibration. Changing the library afterward changes the predictor and requires recalibration. The evidence record supports checking executed operations; it does not certify a language model’s internal reasoning.

3.3 CANDIDATE-AWARE SCORE ALIGNMENT

At round r , a frozen generator constructs a finite, nonempty candidate set $C_r(x) \subseteq \mathcal{Y}$ without access to test labels. A deterministic normalization rule merges formatting variants while preserving spatial distinctions such as “left” and “right.” We introduce a distinct residual label $\text{OTHER} \notin \mathcal{Y}$ and define

$$g_{x,r}(y) = \begin{cases} y, & y \in C_r(x), \\ \text{OTHER}, & y \notin C_r(x). \end{cases} \quad (2)$$

The residual category is scored explicitly through a forced-choice “none of these” option or a frozen sample-frequency rule. Its probability is not assumed to be zero. All experts score the same augmented set.

Let $z_{j,r}(a)$ denote the critic-checked score of expert $j \in \{e, g\}$ for answer a , produced by a scoring rule fixed during development. When an API does not expose token probabilities, a prespecified sampling procedure provides smoothed frequencies whose logarithms serve as scores. Its randomness distribution and sample count are part of the frozen policy. For each expert, fit a scalar temperature $T_j > 0$ on development scenes by minimizing the negative log likelihood of the mapped label in Eq. (2), pooled across the prespecified rounds. Then compute

$$p_{j,r}(a | x) = \frac{\exp(z_{j,r}(a)/T_j)}{\sum_{b \in C_r(x) \cup \{\text{OTHER}\}} \exp(z_{j,r}(b)/T_j)}, \quad \bar{p}_r(a | x) = \frac{1}{2} [p_{e,r}(a | x) + p_{g,r}(a | x)]. \quad (3)$$

Temperature scaling adjusts score sharpness before fusion. It does not ensure that fused scores equal conditional correctness probabilities. The finite-sample guarantee comes from calibrating the final nonconformity score $s_r(x, a) = 1 - \bar{p}_r(a | x)$ and does not require independent experts.

3.4 SCENE-LEVEL CALIBRATION AND BOUNDED ACQUISITION

Questions from the same scene share geometry and perceptual errors, so calibration uses scenes as exchangeable blocks. Let $\mathcal{D}_{\text{cal}}$ contain $m \geq 1$ scenes with finite, nonempty question sets Q_i selected by a prespecified protocol. Before calibration, fix a total risk budget $0 < \alpha < 1$ and round allocations $\alpha_r > 0$ satisfying $\sum_{r=0}^B \alpha_r \leq \alpha$. Every calibration question executes all $B + 1$ potential rounds, including rounds after a possible early acceptance. Compute one maximum score per scene and round:

$$S_{i,r} = \max_{q \in Q_i} s_r(x_{iq}, g_{x_{iq},r}(y_{iq})), \quad k_r = \lceil (m + 1)(1 - \alpha_r) \rceil. \quad (4)$$

The threshold $\hat{q}_r$ is the k_r th order statistic of $S_{1,r}, \dots, S_{m,r}, +\infty$. Thus, an insufficient calibration sample yields an infinite threshold rather than an invalid quantile. The answer set is

$$\Gamma_r(x) = \{a \in C_r(x) \cup \{\text{OTHER}\} : s_r(x, a) \leq \hat{q}_r\}. \quad (5)$$

At inference, accept only a singleton $\Gamma_r(x) = \{a\}$ with $a \neq \text{OTHER}$. Otherwise acquire the next scheduled evidence if $r < B$, and abstain when the budget is exhausted. Empty sets and residual singletons follow the same rule. For evaluation that requires an answer on every question, forced-answer mode returns $\arg \max_{a \in C_r(x)} \bar{p}_r(a | x)$ at the accepted round, or at round B after abstention, using a fixed tie-breaking rule. This mode has no conformal accuracy guarantee; omission of the true candidate necessarily causes an error.

The full potential path may depend on observations through a frozen policy, but not on fitted conformal thresholds. Those thresholds only decide whether to stop. Candidate generation, prompts, model versions, tool library, score transformations, and acquisition budgets all remain fixed after development. This condition makes the complete path well defined for calibration and test scenes alike.

Proposition 1 (Simultaneous coverage and adaptive stopping). *Conditional on the development data and all choices fixed before calibration, suppose the m calibration scenes and one test scene are exchangeable, including their question sets and policy randomness. Then the construction above satisfies*

$$\Pr[\forall q \in Q_{\text{test}}, \forall r \in \{0, \dots, B\} : g_{x_q,r}(y_q) \in \Gamma_r(x_q)] \geq 1 - \alpha. \quad (6)$$

Consequently, the probability of at least one incorrect automatically accepted answer in the test scene is at most α , even when each question stops at a different data-dependent round.

The proof appears in Appendix A. The guarantee is marginal over a new scene, not conditional on a particular room, question type, or acceptance event. In particular, $\Pr(\text{wrong} \mid \text{answered})$ is not bounded by α in general. Retaining OTHER acknowledges a missing candidate but does not identify the missing string. Distribution shifts in pose, visibility, scene composition, or foundation-model behavior can break exchangeability and invalidate the bound.

Cost. With $J = 2$ experts and at most K augmented candidates per round, inference requires at most $J(B + 1)$ expert evaluations and $O(JK(B + 1))$ score processing. A sampled-score evaluation may itself require multiple model calls. The experiments measure additional observations as acquisition cost; model tokens, tool execution, rendering, and wall-clock latency require separate measurements. Template reuse can reduce repeated program synthesis, but its runtime benefit depends on the implementation and workload.

4 EXPERIMENTS

4.1 SYNTHETIC 3D DATA AND EXPERIMENTAL SETUP

We evaluate the uncertainty controller on synthetic 3D data with exact geometric ground truth. Each scene contains 12 object centroids drawn from four categories and an observer with a specified position and orientation. One question per independent scene asks for a category count in front of the observer, a target object’s dominant relative direction, or the nearest object’s category. Counts above five share the label “6+.” With probability 0.20, candidate generation removes one uniformly sampled entry from the query’s answer vocabulary, independently of the true answer. This procedure yields a clean-test candidate recall of $95.60\% \pm 0.27\%$ across seeds; the removal probability is distinct from the probability of omitting the true answer.

Two geometric estimators instantiate the expert scoring branches. The egocentric estimator has per-observation object retention 0.60, position-noise standard deviation 0.25 m, and yaw-noise standard deviation 0.15 rad. The global estimator uses 0.82, 0.45 m, and 0.32 rad, respectively, providing more complete but less precise observations. Each acquisition supplies one additional observation to both estimators. Coordinates and yaw errors are averaged across observations, and 64 Monte Carlo draws form each answer histogram. Object categories and observer translation are known. Raising the smoothed probabilities to powers 2.2 and 0.65, followed by normalization, creates a controlled mismatch in confidence sharpness.

For each of five seeds, the dataset comprises 400 development, 600 calibration, 1,000 clean-test, and 1,000 shifted-test scenes. Partitions have disjoint identifiers and separate random streams. We select one temperature per expert by development-set negative log likelihood over 81 logarithmically spaced values from 0.25 to 4. The total risk budget is $\alpha = 0.10$. Fixed-round methods allocate the entire budget to one round; adaptive inference allocates $\alpha/3$ to each of rounds 0, 1, 2. All potential calibration rounds are completed before fitting the thresholds. With one question per scene, scene-maximum calibration reduces to ordinary split conformal calibration. Appendix C gives the data-generation and observation specifications.

Compared methods. Egocentric, Global, Unscaled fusion, and Aligned fusion produce point answers using the initial observation. Unscaled fusion averages the two unaligned expert distributions; Aligned fusion applies temperature scaling before averaging. Conformal fixed (0) uses the aligned initial scores, and Conformal fixed (2) uses all three observations. Adaptive aligned applies the stopping rule in Section 3.4 with at most two additional acquisitions.

Metrics. Forced accuracy (Acc.) evaluates the highest-scoring ordinary candidate on every question, including those on which the controller abstains. Mapped-label coverage (Cov.) measures inclusion of $g_{x,r}(y)$ in the prediction set, including OTHER. Size is the mean number of retained labels. Answer rate (Ans.) is the fraction of non-residual singleton outputs; selective accuracy (Sel.) is accuracy among those outputs. Joint error is the fraction of all questions answered incorrectly. Adaptive metrics use the first accepted round, or round 2 after abstention. For point baselines, the reported set is the predicted ordinary singleton, so coverage and selective accuracy equal forced accuracy. Results are means and sample standard deviations over five seeds; they are not confidence intervals.

4.2 ALIGNMENT, ABSTENTION, AND ACQUISITION

Table 1 shows that alignment raises forced accuracy from 72.80% to 74.80%, a gain of 2.00 percentage points. The strongest individual estimator, Global, reaches 72.30% at the initial round. These results support aligning expert distributions before fusion under the specified confidence mismatch.

Table 1: Clean-test results on synthetic 3D data: 1,000 scenes per seed and five seeds. Entries are mean $\pm$ sample standard deviation. All metrics except Size are percentages. Fixed (r) uses round r ; adaptive inference uses at most two additional acquisitions.

Method	Acc. (%)	Cov. (%)	Size	Ans. (%)	Sel. (%)
Egocentric	68.8 ± 1.0	68.8 ± 1.0	1.00 ± 0.00	100.0 ± 0.0	68.8 ± 1.0
Global	72.3 ± 0.9	72.3 ± 0.9	1.00 ± 0.00	100.0 ± 0.0	72.3 ± 0.9
Unscaled fusion	72.8 ± 0.9	72.8 ± 0.9	1.00 ± 0.00	100.0 ± 0.0	72.8 ± 0.9
Aligned fusion	74.8 ± 0.9	74.8 ± 0.9	1.00 ± 0.00	100.0 ± 0.0	74.8 ± 0.9
Conformal fixed (0)	74.8 ± 0.9	90.8 ± 0.6	1.50 ± 0.03	59.6 ± 1.6	93.5 ± 0.6
Conformal fixed (2)	89.6 ± 0.8	92.5 ± 0.6	1.11 ± 0.03	92.0 ± 1.0	95.3 ± 0.4
Adaptive aligned	87.1 ± 1.0	96.7 ± 0.5	1.22 ± 0.03	83.6 ± 1.7	97.4 ± 0.5

Table 2: Shifted-test results on synthetic 3D data with clean-data temperatures and thresholds: 1,000 scenes per seed and five seeds. The nominal 90% coverage target is not guaranteed under shift. Metrics, units, and aggregation match Table 1.

Method	Acc. (%)	Cov. (%)	Size	Ans. (%)	Sel. (%)
Aligned fusion	60.6 ± 1.2	60.6 ± 1.2	1.00 ± 0.00	100.0 ± 0.0	60.6 ± 1.2
Conformal fixed (0)	60.6 ± 1.2	84.2 ± 0.8	1.79 ± 0.04	41.6 ± 1.8	79.4 ± 0.8
Conformal fixed (2)	80.0 ± 1.0	82.8 ± 1.1	1.01 ± 0.02	89.8 ± 1.4	83.1 ± 0.9
Adaptive aligned	76.6 ± 1.1	88.2 ± 0.8	1.34 ± 0.03	78.0 ± 1.5	85.9 ± 0.7

Conformal calibration changes acceptance while preserving the underlying score ranking. At round 0, it answers 59.60% of questions with 93.48% selective accuracy and 90.80% mapped-label coverage. The coverage exceeds the nominal 90% target in this experiment. The theoretical bound is marginal over calibration and test scenes; it does not require every realized test sample to reach that target.

Adaptive inference answers 83.60% of questions with 97.39% selective accuracy, 96.70% mapped-label coverage, and 2.18% joint error. It uses 1.06 additional acquisitions on average, compared with 2.00 for Conformal fixed (2), a reduction of 46.80% (Table 4). The final fixed-round method achieves higher forced accuracy (89.60% versus 87.10%) and a higher answer rate (92.00%), with lower selective accuracy (95.28%). Adaptive inference therefore trades some answer availability for fewer observations and more accurate accepted answers. Both early stopping and the stricter per-round risk allocation contribute to this comparison; it does not isolate the effect of early stopping at matched coverage or answer rate.

4.3 SHIFT STRESS TEST AND LIMITS OF CALIBRATION

We retain the clean-data temperatures and thresholds while reducing each estimator’s retention probability by 0.20, multiplying position-noise standard deviations by 1.7, and multiplying yaw-noise standard deviations by 2.5. The estimators use the shifted noise scales in their Monte Carlo draws, but temperatures and conformal thresholds remain unchanged. This experiment tests calibration transfer under a specified observation shift.

Adaptive mapped-label coverage falls to 88.16% and selective accuracy to 85.89% (Table 2). Coverage decreases by 8.54 percentage points, while joint error rises to 11.00%, above the 10% clean-distribution risk budget. Figure 2 also shows lower coverage and answer rates for both fixed-round rules. Additional observations improve forced accuracy under shift, but do not restore the exchangeability required by the clean-data thresholds. The results demonstrate the need to assess distribution change before transferring a calibration guarantee.

5 DISCUSSION AND LIMITATIONS

The experiments establish controller behavior on synthetic 3D data with known object identities and translation, independent missingness, and Gaussian observation noise. They evaluate score align-

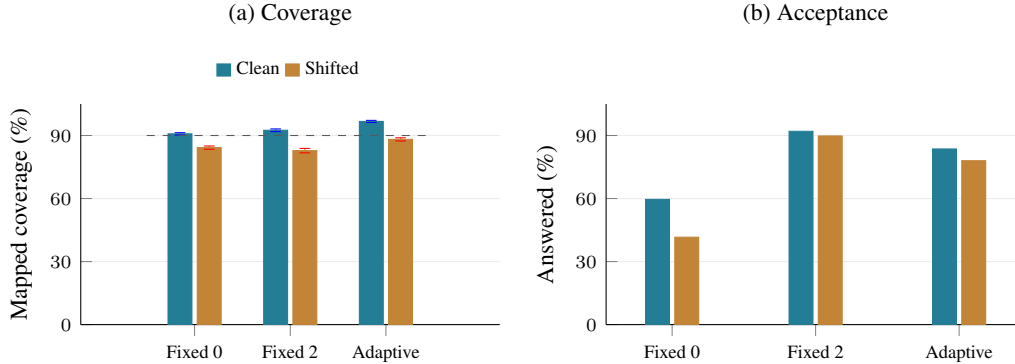


Figure 2: Coverage and answer rate under observation shift. Bars show means over five seeds; coverage error bars show one sample standard deviation. The dashed line marks the nominal 90% coverage target. The methods use different acquisition rules and per-round risk allocations.

ment, conformal acceptance, and acquisition, while the vision-language experts, generated-program library, and language-model critic remain architectural components beyond this empirical study. SQA3D evaluation would additionally require learned perception, natural-language grounding, explicitly declared pose and instance inputs, and official answer matching. Synthetic acquisition counts cannot establish foundation-model latency or token savings.

The theoretical guarantee requires labeled calibration data and conditional exchangeability. Although it allows multiple correlated questions per scene, the experiments use one question per scene and therefore do not measure the conservatism of scene maxima in that setting. Splitting risk across rounds can increase abstention, and candidate recall limits forced accuracy. The residual label identifies unresolved candidate uncertainty without recovering the missing answer. For deployment, changes to perception, prompts, geometry tools, or acquisition policy require reassessing calibration. In robotic and accessibility settings, accepted QA outputs also require a separate controller for safe physical actions.

6 CONCLUSION

TAME3D connects evidence-based multi-agent reasoning with calibrated decisions about when to answer. Its design combines typed geometric operations, aligned expert scores, and candidate-aware conformal sets. Scene-level calibration and a fixed allocation of risk across rounds preserve a finite-sample error bound under adaptive stopping. On synthetic 3D data, alignment improves forced accuracy, while adaptive acquisition achieves 97.39% selective accuracy with 1.06 additional acquisitions on average. This saving comes with a lower answer rate than final-round inference. The coverage loss of 8.54 percentage points under observation shift further shows why acquisition and calibration must be evaluated together. Extending this controller to learned perception and natural-language grounding, and improving efficiency for scenes with many correlated questions, are promising directions.

REPRODUCIBILITY

The source package includes the experimental summary counts, the scripts that generate all numerical tables and plot data, editable vector figures, and compilation instructions. Appendix C specifies the synthetic data and observation models, random seeds, and metric aggregation. The package supports regeneration of the reported aggregate statistics and manuscript; it does not include the simulator or per-scene prediction traces.

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A COVERAGE PROOF AND INTERPRETATION

Condition on the development data and all choices fixed before calibration, as in Proposition 1. At a fixed round r , applying the same frozen predictor to each scene preserves exchangeability of the m calibration maxima and the test-scene maximum. The split-conformal rank argument, with ties handled conservatively by a non-strict threshold, gives

$$\Pr(S_{\text{test},r} > \hat{q}_r) \leq \alpha_r. \quad (7)$$

When $k_r = m + 1$, the threshold is infinity and the event is impossible. Otherwise, the rank bound is $(m + 1 - k_r)/(m + 1) \leq \alpha_r$. Every question score in a test scene is at most the scene maximum, so failure to cover any mapped label at round r is exactly the event $S_{\text{test},r} > \hat{q}_r$. A union bound over the $B + 1$ rounds yields Eq. (6). Independence between experts, rounds, or questions within a scene is unnecessary.

On the event of simultaneous coverage, any accepted singleton excluding OTHER must equal the mapped true label. Because a non-residual mapped label is the original canonical answer itself, such an accepted answer is correct. A stopping rule that selects among these rounds cannot introduce an error on that event. Thus the probability of at least one wrong accepted answer is bounded by the union of the coverage failures.

This result does not give a conditional error bound for the subset of accepted questions. For a single prespecified question and nonzero acceptance probability, it only implies

$$\Pr(\text{wrong} \mid \text{accepted}) \leq \min \left\{ 1, \frac{\alpha}{\Pr(\text{accepted})} \right\}. \quad (8)$$

Empirical selective accuracy is therefore reported rather than asserted to be at least $1 - \alpha$. Scene-level maxima can also be conservative when a scene has many questions. The collection and size of a test scene’s question set must follow the same protocol as calibration; arbitrary changes to the question-selection procedure are a distribution shift.

Residual labels and finite sets. The coverage event concerns $g_{x,r}(y)$, not an unknown answer string. If $\text{OTHER} \in \Gamma_r(x)$, one could lift the set to include every string outside $C_r(x)$, but this would not produce a useful finite answer list. Abstention is the operational choice. Report both mapped-label coverage and candidate recall $\Pr(y \in C_r(x))$ to make this distinction visible.

Why every calibration path is completed. Suppose calibration examples with early singleton sets were removed before later thresholds were fitted. Later-round calibration would then condition on a stopping event that depends on estimated thresholds, and the unqualified rank argument would no longer apply. We instead compute all potential round scores on every calibration scene. At deployment, unneeded rounds are skipped only to save acquisition cost.

B INFERENCE PROCEDURE AND INTERFACES

Table 3: Bounded inference after development and calibration. The acquisition policy, score transformations, risk allocation, and thresholds are fixed before test-time execution.

1	Normalize the question and establish the pose regime and coordinate frame.
2	Initialize the evidence ledger and the frozen acquisition policy.
3	For each round $r = 0, \dots, B$, acquire the scheduled views or tool outputs.
4	Generate candidates C_r , append OTHER, and score the same ordered list with both experts.
5	Apply deterministic critic checks, temperature scaling, and mean fusion.
6	Construct Γ_r using its separately calibrated threshold.
7	Return the unique answer and its evidence if Γ_r is a non-residual singleton.
8	After the final round, return an abstention with the unresolved evidence failures.

Expert output schema. Each response contains a question identifier, round number, pose identifier, ordered candidate strings, one real-valued score per candidate including the residual label, cited evidence identifiers, and missing-evidence flags. The interface rejects malformed lengths, non-finite scores, unknown object IDs, and invalid frames. A failed expert receives equal logits over the shared augmented set and a failure flag, producing a uniform distribution after temperature scaling. The same fallback applies during development, calibration, and inference. Frequency-based scores use fixed positive smoothing before taking logarithms.

Role specification. The manager identifies the spatial reference frame and selects operations from the typed library within the declared budget. The egocentric expert restricts visual claims to supplied views and cites visible object IDs. The global expert cites transformed coordinates or scene-graph edges for metric claims. The critic checks these citations and returns structured validity flags. Candidate generation precedes these checks, and the final answer follows the conformal decision rule.

Fixed critic transformation. A compatible critic transformation sets an unsupported ordinary candidate’s score to the minimum finite input score minus a fixed margin. This minimum is computed before modifying any scores. The residual score remains available because failure to establish a relation does not prove its negation. Temperature scaling and softmax then operate on the entire augmented list. The margin and repair rules are fixed on development scenes and replayed unchanged during calibration and inference; total parser failure invokes the uniform fallback. The synthetic experiments use well-formed geometric outputs and do not evaluate a language-model critic.

Worked geometric example. Consider an agent at the origin with identity rotation, a chair at $(-1, 2, 0)$, and a table at $(1, 2, 0)$. The chair has negative local x and lies to the agent’s left; both objects have positive local y and lie in front. An egocentric view that misses the chair cannot establish that the left region is empty. The global expert can support “chair” through the corresponding instance record and transformed coordinates. This example illustrates how the evidence interface distinguishes missing observations from verified spatial relations.

C SYNTHETIC DATA, METRICS, AND ADDITIONAL RESULTS

Geometry and randomness. Each object’s horizontal coordinates are uniform on $[-4, 4]^2$ m and its height is uniform on $[0, 2.5]$ m. Categories are independent uniform draws from four symbols. The observer’s horizontal position is uniform on $[-1, 1]^2$ m, with height 1.4 m and yaw $\theta \sim \text{Uniform}[-\pi, \pi]$. The simulator defines forward as $(\cos \theta, \sin \theta, 0)$ and right as $(\sin \theta, -\cos \theta, 0)$. These are the second and first columns of R in Eq. (1); therefore yaw zero faces world $+x$ and is not the identity rotation used in the separate worked example.

The query type is uniform among count, direction, and nearest-category queries. A count query draws a category uniformly and counts its centroids with positive forward coordinate. A direction query draws one of the 12 target indices uniformly and selects the dominant horizontal axis; ties favor the forward/backward axis. A nearest query uses Euclidean distance in three dimensions. Each scene receives one query only. Seeds are 17, 29, 43, 71, 101, and each scene uses a NumPy random stream keyed by seed, partition identifier, and scene index. The candidate mask is sampled before the answer is computed and stays fixed across all rounds.

Observation and score construction. Visibility is an independent Bernoulli retention event for each object and each estimator at every round. An observed centroid receives isotropic Gaussian position noise; the observer’s yaw receives an independent Gaussian error. Coordinates are averaged over observations of that object. The yaw error is averaged over all available views. For each predictive draw, perturb the averaged coordinates by the estimator’s position standard deviation divided by the square root of its observation count, and perturb the averaged yaw analogously. Missing objects remain absent. If the requested direction target is missing, or all objects are missing, use a uniform histogram. These Monte Carlo scores are approximate predictive distributions, not Bayesian posteriors under a complete scene model.

Add 0.5 to every histogram bin before normalization. Sum the mass of removed candidates into OTHER and add residual mass 0.001 before applying the confidence power and renormalizing.

Known categories and known translation intentionally isolate missingness, metric noise, yaw uncertainty, and score mismatch. The study does not simulate recognition errors, missing geometry correlated with object size, or common foundation-model hallucinations.

Table 4: Acquisition cost and joint error on synthetic 3D data. Acq. counts additional observations supplied to both experts, excluding the initial observation. Err. is the percentage of all questions answered incorrectly, with abstentions included in the denominator. Entries are mean $\pm$ sample standard deviation over five seeds, with 1,000 scenes per condition and seed.

Method	Clean acq.	Shift acq.	Clean err. (%)	Shift err. (%)
Conformal fixed (0)	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	3.88 $\pm$ 0.23	8.54 $\pm$ 0.05
Conformal fixed (2)	2.00 $\pm$ 0.00	2.00 $\pm$ 0.00	4.34 $\pm$ 0.36	15.16 $\pm$ 0.57
Adaptive aligned	1.06 $\pm$ 0.04	1.39 $\pm$ 0.04	2.18 $\pm$ 0.33	11.00 $\pm$ 0.37

The clean adaptive controller’s mean forced accuracies by query type are 85.50% for count, 92.24% for direction, and 83.51% for nearest category. Under the specified shift they become 66.97%, 85.88%, and 77.06%. These are unweighted means of per-seed type-specific accuracies; pooled accuracy need not equal their unweighted average because sampled type counts differ. All-round simultaneous mapped coverage is 93.80% on clean scenes and 83.40% under shift, distinct from the selected-round coverage in the main tables.

Metric aggregation. For each seed, forced accuracy divides the number of correct forced answers by the number of questions. Coverage divides the number of covered mapped labels by the same denominator. Answer rate divides the number of accepted answers by all questions, while selective accuracy divides the number of correct accepted answers by accepted answers only. Set size and acquisition cost divide their respective totals by the number of questions. Each rate is computed per seed before taking its arithmetic mean and sample standard deviation across the five seeds.

Joint error is computed directly from the wrong-and-answered count divided by all questions. Within a seed, this equals the answer rate multiplied by one minus selective accuracy, when both rates are expressed as fractions. The identity does not generally hold after averaging the two factors separately across seeds. Percentage-point differences and acquisition savings use unrounded means; in particular, the reported 46.80% saving is calculated before rounding the displayed acquisition counts.

Consistency checks. The analysis script checks count bounds, candidate recall, set-size and acquisition budgets, and agreement between overall and query-type correctness. Tables, plot data, and numerical statements are generated from a single summary file. The supplementary package therefore permits independent recomputation of aggregate statistics. Re-executing the geometric experiment additionally requires the simulator and scene-level outputs.